Random Forest. Example.

Name:________________________SBU ID:________________________

Please include (1) Rmd file; (2) Output from Rmd with answers to all the questions asked

Random Forest with the Banknote Data – Classification Task

The banknote.csv data (see attached) were extracted from images that were taken from genuine and forged banknote-like specimens. Yes, this is a ***Catch Me if You Can*** story. For digitization, an industrial camera usually used for print inspection was used. The final images have 400x 400 pixels. Wavelet Transform tool were used to extract features from images. **There are 1372 banknotes, and 5 variables:**

1. variance of Wavelet Transformed image (continuous)
2. skewness of Wavelet Transformed image (continuous)
3. curtosis of Wavelet Transformed image (continuous)
4. entropy of image (continuous)
5. class (binary) - this is the response variable of interest, 1 (forged) or 0 (genuine).

Your task is to split the data randomly into training (75%) and testing (25%), first build the best random forest to predict 'class' using the training data, then use the out-of-bag (OOB) data to measure its performance, and then use that model to predict whether each check in the testing data is forged or genuine. Please use *randomforest* function in R to build the random forest classifier.

Please review the following website for related methods and concepts:
http://www.sthda.com/english/articles/35-statistical-machine-learning-essentials/140-bagging-and-random-forest-essentials/

1. Please use the random seed 123 to divide the cleaned data into 75% training and 25% testing.

2. Please first build the best random forest to predict 'class' using the training data. Please compute the Confusion matrix and report the sensitivity (that is, a forged check is found to be forged), specificity (that is, a genuine check is found to be genuine), and the overall accuracy using the out of bag (OOB) samples.

3. Next please use this random forest to predict 'class' in the testing data. Please compute the Confusion matrix and report the sensitivity, specificity and the overall accuracy for the testing data.

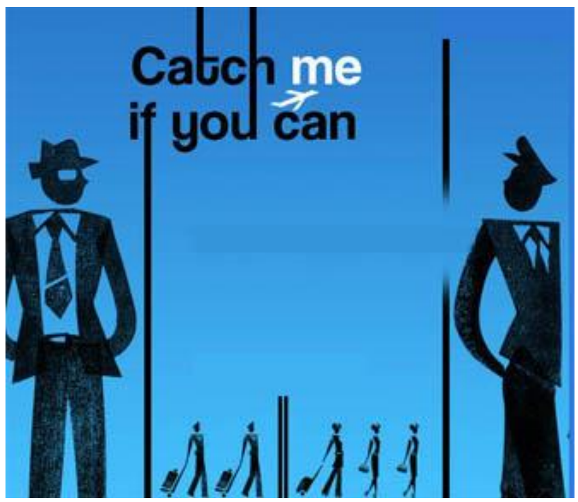
4. Please plot the variables importance measures using
    a. *MeanDecreaseAccuracy*, which is the average decrease of model accuracy in predicting

the outcome of the out-of-bag samples when a specific variable is excluded from the model.

b. *MeanDecreaseGini*, which is the average decrease in node impurity that results from splits over that variable. The Gini impurity index is only used for classification problem.

5. Please show the importance of each variable in percentage based on *MeanDecreaseAccuracy*.

6. In a classification task using the random forest, suppose we have 25 variables (as predictors) in the original data set – then at each node split, what is the number of variables we should (as commonly recommended) to select, at random, to be considered for that node split?



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